



In []:

```
In [76]: import pandas as pd
import seaborn as sns
import numpy as np
from apyori import apriori
```

```
In [77]: df=pd.DataFrame(pd.read_csv('Market_Basket - Market_Basket.csv',header=None))
```

In [78]: df

```
Out[78]:
```

	0	1	2	3	4	5	6	7	8
0	shrimp	almonds	avocado	vegetables mix	green grapes	whole weat flour	yams	cottage cheese	eggs
1	burgers	meatballs	eggs	NaN	NaN	NaN	NaN	NaN	NaN
2	chutney	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	turkey	avocado	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	mineral water	milk	energy bar	whole wheat rice	green tea	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...
7496	butter	light mayo	fresh bread	NaN	NaN	NaN	NaN	NaN	NaN
7497	burgers	frozen vegetables	eggs	french fries	magazines	green tea	NaN	NaN	NaN
7498	chicken	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
7499	escalope	green tea	NaN	NaN	NaN	NaN	NaN	NaN	NaN
7500	eggs	frozen smoothie	yogurt cake	low fat yogurt	NaN	NaN	NaN	NaN	NaN

7501 rows × 20 columns

```
In [79]: df.shape
```

Out[79]: (7501, 20)

```
In [80]: df_list=(df.to_numpy())
df_list
```

```
Out[80]: array([[ 'shrimp', 'almonds', 'avocado', ..., 'frozen smoothie',
                'spinach', 'olive oil'],
               [ 'burgers', 'meatballs', 'eggs', ..., nan, nan, nan],
               [ 'chutney', nan, nan, ..., nan, nan, nan],
               ...,
               [ 'chicken', nan, nan, ..., nan, nan, nan],
               [ 'escalope', 'green tea', nan, ..., nan, nan, nan],
               [ 'eggs', 'frozen smoothie', 'yogurt cake', ..., nan, nan, nan]],
          dtype=object)
```

```
In [ ]:
```

```
In [81]: #Use Series
x=pd.Series(df_list.flatten())
x.dropna(inplace=True)
```

```
In [82]: sns.barplot(x=x.value_counts().head(20).index,y=x.value_counts().head(20),palette
```

C:\Users\ARJUN\AppData\Local\Temp\ipykernel_66352\1267659924.py:1: FutureWarning:

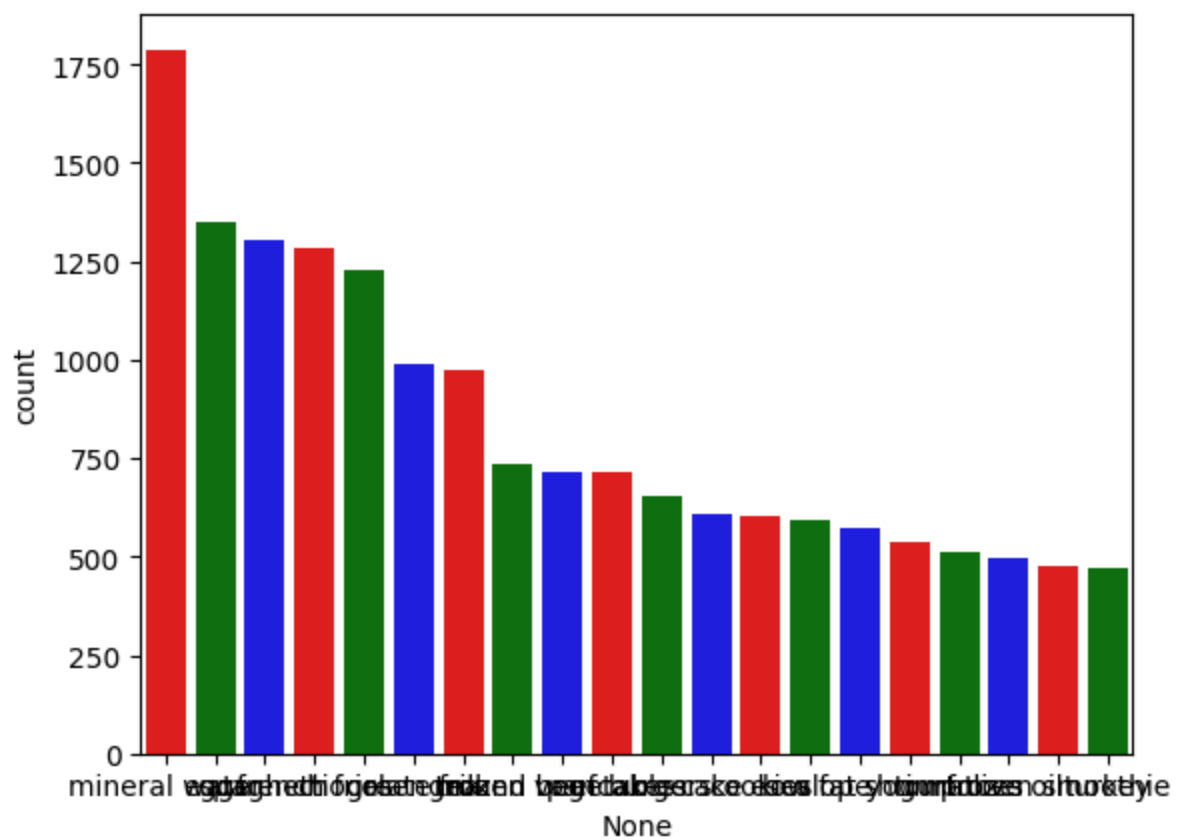
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=x.value_counts().head(20).index,y=x.value_counts().head(20),palette=['red','green','blue'])
```

C:\Users\ARJUN\AppData\Local\Temp\ipykernel_66352\1267659924.py:1: UserWarning: The palette list has fewer values (3) than needed (20) and will cycle, which may produce an uninterpretable plot.

```
sns.barplot(x=x.value_counts().head(20).index,y=x.value_counts().head(20),palette=['red','green','blue'])
```

```
Out[82]: <Axes: xlabel='None', ylabel='count'>
```



```
In [83]: df_list=df_list.tolist()
for ele in df_list:
    while np.nan in ele:
        ele.remove(np.nan)
df_list
```

```
Out[83]: [['shrimp',
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```

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'soup',

'olive oil'],
['burgers',
'shrimp',
'mineral water',
'chicken',
'fromage blanc',
'salmon',
'eggs',
'cake',
'frozen smoothie',
'strawberries'],
['green tea'],
['cookies'],
['french fries', 'cookies'],
['milk', 'butter', 'eggs'],
['eggs', 'mushroom cream sauce', 'low fat yogurt'],
['eggs', 'green tea'],
['mineral water', 'whole wheat rice'],
['shrimp',
'frozen vegetables',
'parmesan cheese',
'mineral water',
'whole wheat rice',
'cake'],
['whole wheat rice', 'pancakes'],
['frozen vegetables',
'spaghetti',
'olive oil',
'butter',
'salmon',
'oil',
'cooking oil',
'frozen smoothie',
'cauliflower'],
['green tea'],
['fresh tuna', 'eggs', 'escalope', 'strawberries'],
['spaghetti', 'soup', 'milk', 'carrots', 'chocolate'],
['burgers', 'mineral water', 'soup', 'meatballs', 'olive oil'],
['grated cheese',
'mineral water',
'cooking oil',
'french fries',
'cookies',
'low fat yogurt'],
['shrimp', 'milk', 'olive oil', 'french fries'],
['frozen vegetables', 'mineral water', 'pancakes', 'cake', 'tomato juice'],
['spaghetti', 'milk', 'chicken'],
['grated cheese',
'mineral water',
'chicken',
'french fries',
'cottage cheese',
'pancakes'],
['ground beef', 'mineral water', 'milk', 'eggs', 'mint'],

```

['shrimp', 'body spray', 'green tea'],
['frozen smoothie'],
['herb & pepper', 'frozen vegetables', 'mineral water', 'muffins', 'cereal
s'],
['turkey',
 'tomatoes',
 'spaghetti',
 'milk',
 'cider',
 'eggs',
 'honey',
 'cake',
 'green tea',
 'french fries',
 'brownies',
 'tomato juice'],
...]

```

```

In [84]: rules1=apriori(df_list,min_support=0.05,min_confidence=0.05)
r1=list(rules1)

```

```

In [85]: frequent_items = []
supports = []
confidences = []
lifts = []

```

```

In [86]: for r in r1:
    if not hasattr(r, "ordered_statistics") or len(r.ordered_statistics) == 0:
        continue
    for x in r.ordered_statistics:
        if len(x.items_base) > 0 and len(x.items_add) > 0:
            frequent_items.append(tuple(x.items_base) + tuple(x.items_add))
            supports.append(r.support)
            confidences.append(x.confidence)
            lifts.append(x.lift)

```

```

In [87]: min_len = min(len(frequent_items), len(supports), len(confidences), len(lifts))
frequent_items = frequent_items[:min_len]
supports = supports[:min_len]
confidences = confidences[:min_len]
lifts = lifts[:min_len]

```

```

In [88]: rules1_df = pd.DataFrame({
    'Frequent Itemset': frequent_items,
    'Support': supports,
    'Confidence': confidences,
    'Lift': lifts
}).sort_values(by='Lift', ascending=False).reset_index(drop=True)
rules1_df

```

Out[88]:

	Frequent Itemset	Support	Confidence	Lift
--	------------------	---------	------------	------

0	(mineral water, spaghetti)	0.059725	0.250559	1.439085
1	(spaghetti, mineral water)	0.059725	0.343032	1.439085
2	(chocolate, mineral water)	0.052660	0.321400	1.348332
3	(mineral water, chocolate)	0.052660	0.220917	1.348332
4	(mineral water, eggs)	0.050927	0.213647	1.188845
5	(eggs, mineral water)	0.050927	0.283383	1.188845

```
In [89]: rules2=apriori(df_list,min_support=0.01,min_confidence=0.01)
r2=list(rules2)
```

```
In [90]: frequent_items = []
supports = []
confidences = []
lifts = []
```

```
In [91]: for r in r2:
    if not hasattr(r, "ordered_statistics") or len(r.ordered_statistics) == 0:
        continue
    for x in r.ordered_statistics:
        if len(x.items_base) > 0 and len(x.items_add) > 0:
            frequent_items.append(tuple(x.items_base) + tuple(x.items_add))
            supports.append(r.support)
            confidences.append(x.confidence)
            lifts.append(x.lift)
```

```
In [92]: min_len = min(len(frequent_items), len(supports), len(confidences), len(lifts))
frequent_items = frequent_items[:min_len]
supports = supports[:min_len]
confidences = confidences[:min_len]
lifts = lifts[:min_len]
```

```
In [94]: rules2_df = pd.DataFrame({
    'Frequent Itemset': frequent_items,
    'Support': supports,
    'Confidence': confidences,
    'Lift': lifts
}).sort_values(by='Lift', ascending=False).reset_index(drop=True)
rules2_df.head()
```


Out[94]:

	Frequent Itemset	Support	Confidence	Lift
0	(herb & pepper, ground beef)	0.015998	0.323450	3.291994
1	(ground beef, herb & pepper)	0.015998	0.162822	3.291994
2	(spaghetti, mineral water, ground beef)	0.017064	0.285714	2.907928
3	(ground beef, spaghetti, mineral water)	0.017064	0.173677	2.907928
4	(olive oil, spaghetti, mineral water)	0.010265	0.155870	2.609786

In []:

In []: